exploration_datasets

Exploratory Data Analysis of Datasets used

Import required library

```
library(dplyr)
library(gtsummary)
library(kableExtra)
library(ggplot2)
```

Global path variable to the data set

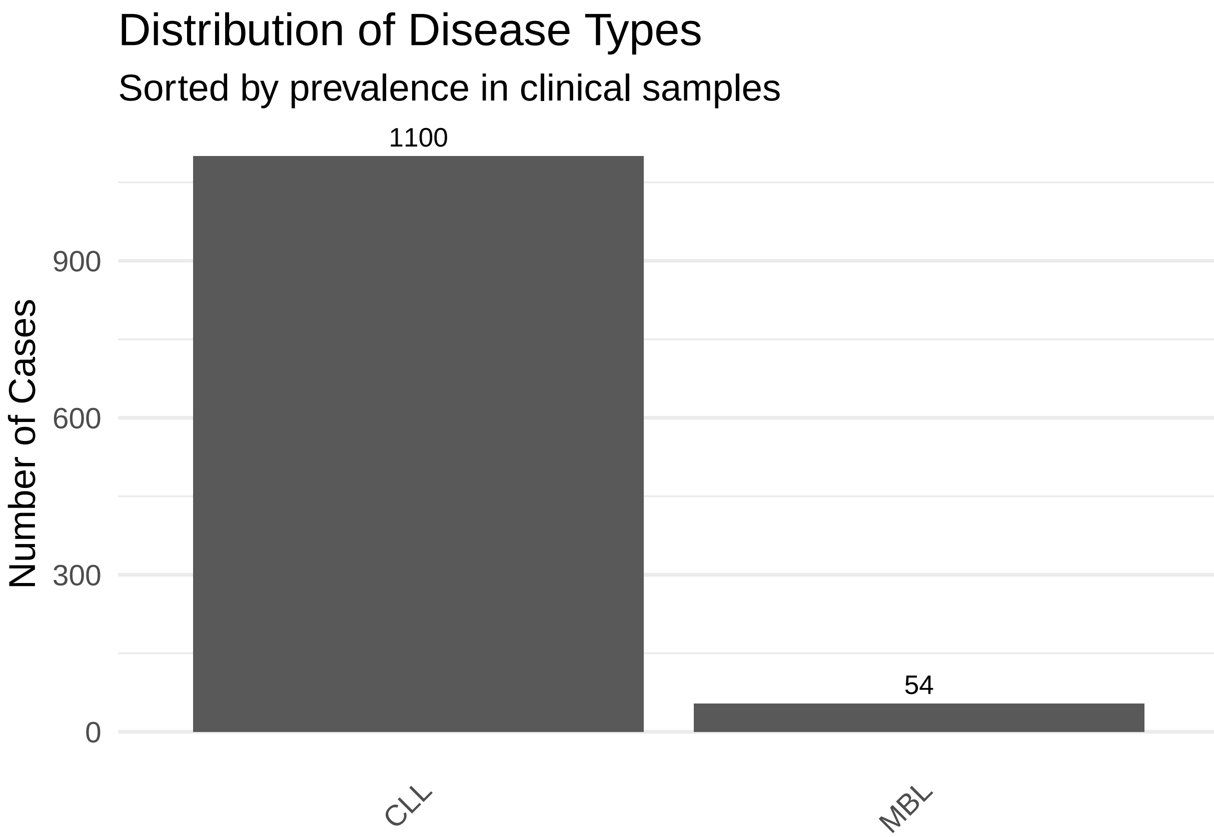
```
dataset_path <- "/Users/sorenbasnet/Documents/Github/STAT_443_FINAL_PROJECT/data/raw/cll_broad_2022"
```

Read in data files

```
data_clinical_patient <- read.table(file.path(dataset_path,"data_clinical_patient.txt"),header=TRUE, sep
data_clinical_sample <- read.table(file.path(dataset_path, "data_clinical_sample.txt"), header=TRUE, sep
data_mutations <- read.table(file.path(dataset_path,"data_mutations.txt"),header=TRUE, sep="\t")
```

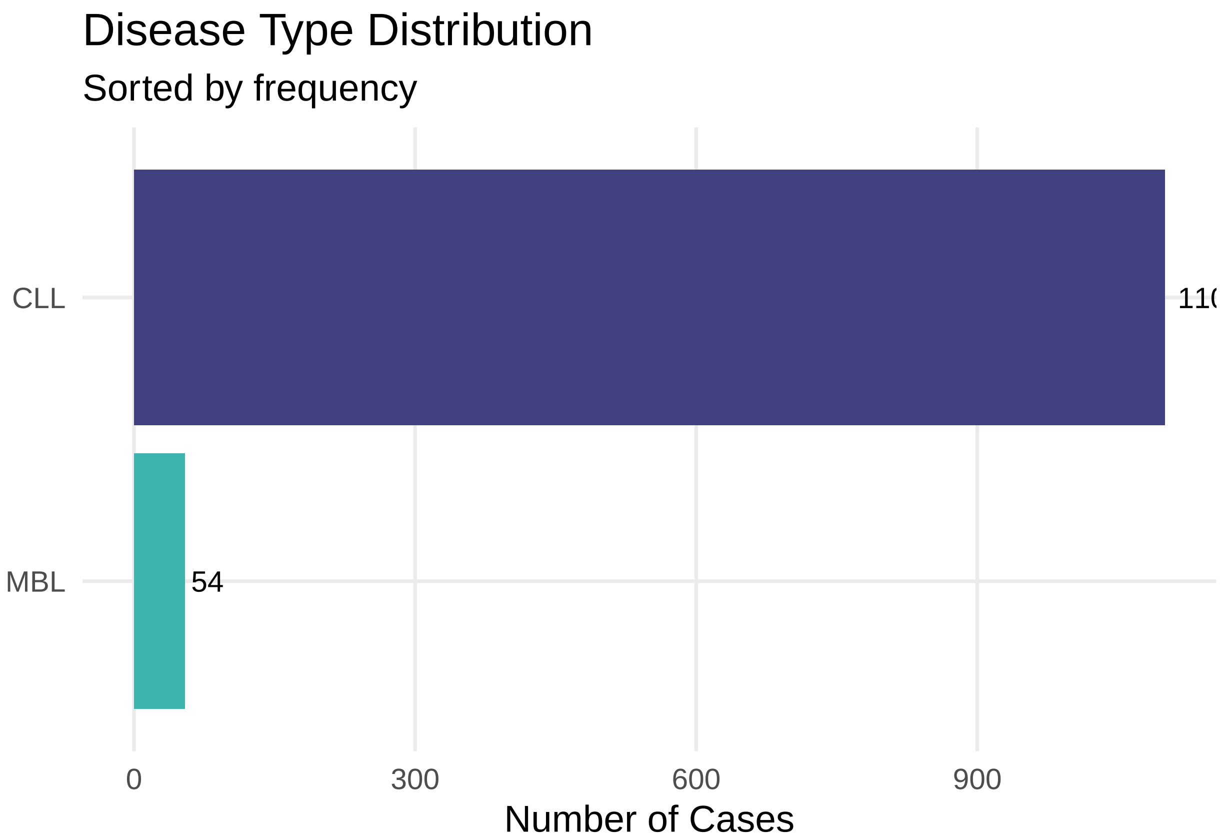
Number of patients with MBL and CLL respectively - data_clinical_sample

```
ggplot(data_clinical_sample, aes(x=DISEASE_TYPE)) +
         geom_bar() +
  geom_text(stat = 'count', aes(label = after_stat(count)), vjust = -0.5, size=3.5) +
  # using a color palette
  scale_fill_viridis_d(option="mako", begin =0.2, end=0.8) +
  theme_minimal(base_size = 14) +
  labs(
    title = "Distribution of Disease Types",
    subtitle = "Sorted by prevalence in clinical samples",
    x = NULL, # Removes the redundant 'DISEASE_TYPE' label
    y = "Number of Cases"
  ) +
  # Clean up the grid and rotate labels if they are long
  theme(
    panel.grid.major.x = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```



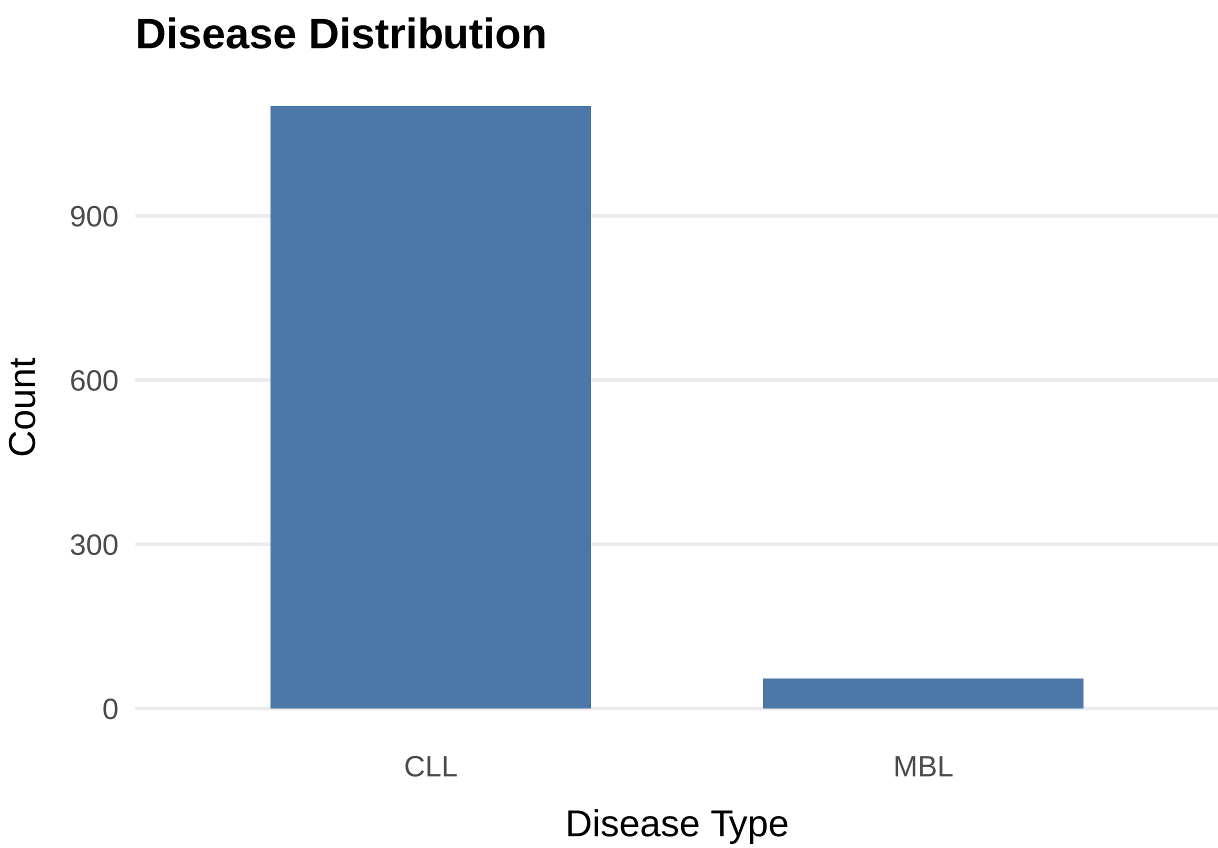

```r
library(ggplot2)
library(forcats) # This is key for the sorting "magic"

ggplot(data_clinical_sample, aes(y = fct_rev(fct_infreq(DISEASE_TYPE)), fill = DISEASE_TYPE)) +
  geom_bar(show.legend = FALSE) +
  # Adds the actual numbers inside/at the end of the bars
  geom_text(stat = 'count', aes(label = after_stat(count)), hjust = -0.2, size = 4) +
  # A sleek, professional color palette
  scale_fill_viridis_d(option = "mako", begin = 0.3, end = 0.7) +
  theme_minimal(base_size = 14) +
  labs(
    title = "Disease Type Distribution",
    subtitle = "Sorted by frequency",
    x = "Number of Cases",
    y = NULL
  ) +
  # Extra clean-up
  theme(panel.grid.minor = element_blank())
```




```r
library(ggplot2)

ggplot(data_clinical_sample, aes(x = DISEASE_TYPE)) +
  geom_bar(fill = "#4C78A8", width = 0.65) +
  labs(
    title = "Disease Distribution",
    x = "Disease Type",
    y = "Count"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    axis.title.x = element_text(margin = margin(t = 10)),
    axis.title.y = element_text(margin = margin(r = 10)),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank()
  )
```



Drawing plots again

```
# 1. Open a PNG file "device" with high resolution (300 DPI)
# Note: I've increased the pixel dimensions to account for the higher resolution
png("disease_plot.png", width = 2400, height = 1800, res = 300)

# 2. Increase margins (6,6,6,6) as you requested
par(mar = c(6, 6, 6, 6))

# 3. Summarize the data
disease_counts <- table(data_clinical_sample$DISEASE_TYPE)

# 4. Basic barplot
my_bar <- barplot(disease_counts,
                  border = FALSE,
                  las = 2,
                  main = "Frequency of Disease Types",
                  col = "skyblue")
# 1. Allow drawing in the margins
par(xpd = TRUE)

# 5. Legend
legend("topright",
       legend = c("CLL", "MBL"),
       fill = "skyblue", # Added fill to match your bars
       bty = "o",
       inset = c(0.00, -0.20))
```

```r
# 6. Save the file
dev.off()
```

```
## pdf
##   2
```

```r
(table(data_clinical_sample$DISEASE_TYPE))
```

```
##
##  CLL  MBL
## 1100   54
```